

Tensor Products, Exterior Products, and Geometric Algebras Aren't Trivial

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Abstract

Tensor, exterior, and geometric algebras can be nicely defined in terms of free algebras with relations. However, this begs the question of whether, once all of the indentifications have been made, what elements remain distinct from zero. This note is a comment on how to check that the identifications don't identify more that one wanted. There is also a comment on understanding the determinant. There is no claim to originality, although no references seem to be immediately available.

1 Introduction

It is well known that tensor product, exterior products, and geometric products can be defined in terms of generators and relations. However, such definitions leave open the *a priori* possibility that the algebra so defined is degenerate due to relationships which are not obvious. For example, if one considers the groups $\mathbf{Z}/2\mathbf{Z}$ and $\mathbf{Z}/3\mathbf{Z}$ as $\mathbf{Z}$ -modules, then

$$\mathbf{Z}/2\mathbf{Z} \otimes_{\mathbf{Z}} \mathbf{Z}/3\mathbf{Z} = 0,$$

that is, all of the elements are identically equal to zero, as all elements of $\mathbf{Z}/2\mathbf{Z}$ are divisible by 3 = 1 (mod 2), but when the 3 moves accross the $\otimes_{\mathbf{Z}}$, it becomes zero mod 3. Thus the only element of the tensor product is zero. Fortunately, this doesn't have for fields like the reals, $\mathbf{R}$. The purpose of this note is to make this obvious.

2 Tensor Algebra

For a vector space V over a field $\mathbf{F}$, let $W = V^*$ be the dual space. Now, for each integer $n > 0$, let X^n the vector space whose elements are lists of n

vectors, so $[w_1, w_2, \dots, w_n]$, $w_i \in W$, with addition and multiplication defined component-wise. Let $X^0 = \mathbf{F}$, and let

$$\mathbf{X} = \bigoplus_{i=0}^{\infty} X^i. \quad (1)$$

Then $\mathbf{X}$ is a vector space of $\mathbf{F}$, and we can consider the algebra of all $\mathbf{F}$ -linear operators from $\mathbf{X}$ to itself, *i.e.*, $\text{Hom}(\mathbf{X}, \mathbf{X})$. Given any $v \in V$, one can associate a linear operator $\hat{v}$ defined by

$$\hat{v}(x) = 0 \quad x \in \mathbf{F} \quad (2)$$

$$\hat{v}([w_1, w_2, \dots, w_n]) = w_1(v) [w_2, \dots, w_n] \quad [w_1, w_2, \dots, w_n] \in X^n \quad (3)$$

Thus each $v \in V$ induces a linear map, and this can be clearly extended to $V \times V$ in a manner which is probably balanced, and for any basis of V , the ordered pairs of vectors give a basis for $V \otimes V$, *etc.* (Maybe a bit more should be said....)

3 Exterior Product

Using the same definitions for X^i and $\mathbf{X}$ as in the section for tensor products, define

$$\tilde{v}(x) = 0 \quad x \in \mathbf{F} \quad (4)$$

$$\begin{aligned} \tilde{v}([w_1, w_2, \dots, w_n]) &= w_1(v) [w_2, \dots, w_n] \\ &\quad - w_2(v) [w_1, w_3, \dots, w_n] \\ &\quad + \dots \\ &\quad + (-1)^{n+1} w_n(v) [w_1, w_3, \dots, w_{n-1}] \end{aligned} \quad [w_1, w_2, \dots, w_n] \in X^n \quad (5)$$

Again, this embeds each $v \in V$ in the algebra $\text{Hom}(\mathbf{X}, \mathbf{X})$ in such a way that

$$\tilde{v}_1 \tilde{v}_2 + \tilde{v}_2 \tilde{v}_1 = 0.$$

Thus that exterior algebra is isomorphically represented by the subalgebra of $\text{Hom}(\mathbf{X}, \mathbf{X})$ generated by the images of the elements of V .

Further, clearly the grading of the algebra is preserved, *etc.*:

$$\bigwedge V = \bigwedge_{i=0}^{\infty} V^i$$

Also note that if multiplying elements of the algebra we don't need to say anything about factors of $n!$.

Note that for each $w \in W = V^*$, this is well defined:

$$\begin{aligned}
& i_w(v_1 \wedge v_2 \wedge \dots \wedge v_k) \\
& = w(v_1)v_2 \wedge \dots \wedge v_k \\
& - w(v_2)v_1 \wedge v_3 \wedge \dots \wedge v_k \\
& \quad + \dots \\
& + (-1)^{k+1}w(v_k)v_1 \wedge v_2 \wedge \dots \wedge v_{k-1}
\end{aligned} \tag{6}$$

3.1 Determinants

If $V = \mathbf{F}^n$, then treating $v_i \in V$ and column vectors and letting $\omega_i \in W = V^*$ be the standard dual basis, then the determinant is

$$|v_1 \ v_2 \ \dots \ v_n| = \tilde{v}_1 \tilde{v}_2 \dots \tilde{v}_n ([\omega_n, \omega_{n-1}, \dots, \omega_1]).$$

Considering the determinant as a signed volume, note that the determinant is actually fixed up to a scalar constant by the two conditions:

1. If v_i is replaced with αv_i , $\alpha \in \mathbf{f}$, then the determinant is changed by a factor of α .
2. If v_i is replaced by $v_i + \alpha v_j$ for $i \neq j$, the value of the determinant does not change; this corresponds to the idea that a shear parallel to one of the faces of a polytope does not change the volume of the polytope.

(Yes, I should say more ...)

4 Geometric Algebra

Proceeding to the geometric algebra, let $g(v, v') \in \mathbf{F}$ be a symmetric bilinear form defined on each pair $v, v' \in V$. So for each $v \in V$, one has $g(v, \cdot) \in V^*$. Consider $\bigwedge V$ as a vector space, for each $v \in V$ one can associate linear endomorphism on $\bigwedge V$ by

$$v^\dagger(v_1 \wedge \dots \wedge v_k) = \pm i_{g(v, \cdot)}(v_1 \wedge \dots \wedge v_k) \pm v \wedge v_1 \wedge \dots \wedge v_k$$

(or something like that... that's all for now).